Practice workbook · Predict how a drug, toxin or disease changes synaptic transmission

Using the synapse model to explain effects of plaques, enzyme inhibitors, curare, alcohol, or too little or too much dopamine or acetylcholine.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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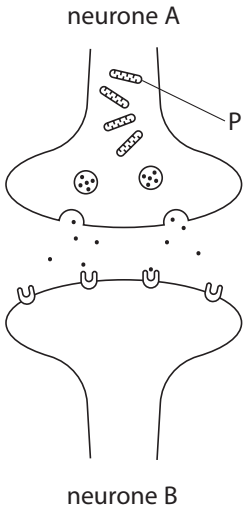
Type ID: HBI-APPLY-SYNAPSE-TO-DRUG-OR-DISEASE

1. Question 8 · part b.ii · 2 marks

November 2020 Paper 2R

Worked-example retry · Attempt part b.ii only.

8 The diagram shows the gap between two neurones, A and B.



(a) (i) Draw an arrow on the diagram to show the direction of travel of an impulse. **(1)**

(ii) Explain the function of structure P in the transmission of the impulse across the gap. **(3)**

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1. Question 8 · part b.ii · 2 marks

November 2020 Paper 2R

Worked-example retry · Attempt part b.ii only.

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(b) Curare is a poison used by South American Indians when hunting. Curare blocks receptors on the post synaptic membranes of neurones supplying skeletal muscles and prevents breathing.

(i) Explain how curare can stop the process of breathing.

(2)

(ii) Acetylcholinesterase is an enzyme found in the gap between two neurones.

Suggest why acetylcholinesterase inhibitors can be used to reverse the effects of curare poisoning.

(2)

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(Total for Question 8 = 8 marks)

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1. Question 8 · part b.ii · 2 marks · Working and answer

Working and answer

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2. Question 6 · part d · 3 marks

June 2019 Paper 2R

Further practice · Attempt part d only.

6 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Alzheimer's disease is a degenerative disease that usually starts slowly and gets worse over time. Scientists want to understand the causes of the disease and discover treatments for the memory loss that is a major feature of Alzheimer's. Scientists use the brains of people who have died to try to find out what happens in the brains of people who have Alzheimer's.

The most common early symptom is short-term memory loss, which is difficulty in remembering recent events. People with this condition are no longer able to transfer information about their daily life from their short-term to long-term memory. For example, they are unable to remember if they went shopping.

The condition causes plaques of an insoluble protein called B-amyloid to build up in the gap between neurones. Over time, these plaques appear in the neurones themselves in the regions of the brain associated with memory.

The B-amyloid is formed when an essential protein called amyloid precursor protein (APP) is not broken down correctly.

(a) Describe two differences between short-term memory and long-term memory (lines 8 to 9). **(2)**

1 .....

.....

2 .....

.....

(b) State the area of the brain responsible for memory. **(1)**

.....

(c) State two symptoms of Alzheimer's disease, other than memory loss. **(2)**

1 .....

.....

2 .....

.....

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P 6 0 1 8 7 A 0 1 6 2 4

2. Question 6 · part d · 3 marks

June 2019 Paper 2R

Further practice · Attempt part d only.

(d) Explain why the build-up of B-amyloid disrupts brain function (lines 10 to 11).

(3)

(e) One method of treating the early stages of Alzheimer's is the use of an enzyme inhibitor.

Explain how this treatment could help people with this condition.

(2)

(f) Describe the evidence that shows the amyloid precursor protein (APP) is involved in the development of Alzheimer's disease (lines 10 to 14).

(2)

(Total for Question 6 = 12 marks)

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2. Question 6 · part d · 3 marks · Working and answer

Working and answer

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3. Question 6 · part e · 2 marks

June 2019 Paper 2R

Further practice · Attempt part e only.

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2 .....

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1 .....

.....

2 .....

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16

3. Question 6 · part e · 2 marks

June 2019 Paper 2R

Further practice · Attempt part e only.

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3. Question 6 · part e · 2 marks · Working and answer

Working and answer

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4. Question 5 · part c.iii · 3 marks

November 2020 Paper 1

Further practice · Attempt part c.iii only.

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5 This question is about the nervous system.

(a) Describe what is meant by a reflex action.

(3)

(b) Diagram 1 shows a reflex arc.

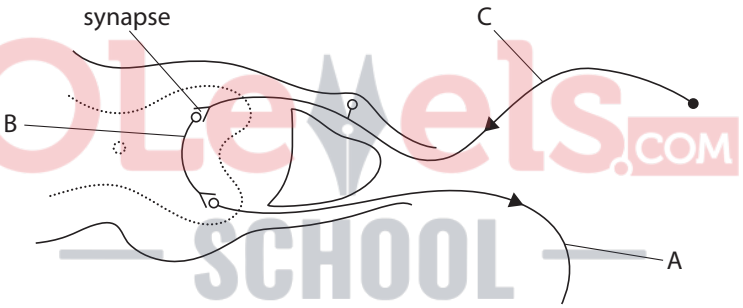


Diagram 1

Name the types of neurone labelled A, B and C.

(3)

A .....

B .....

C .....

4. Question 5 · part c.iii · 3 marks

November 2020 Paper 1

Further practice · Attempt part c.iii only.

(c) Diagram 2 shows a neurone from a healthy brain and a neurone from the brain of someone with Alzheimer's disease.

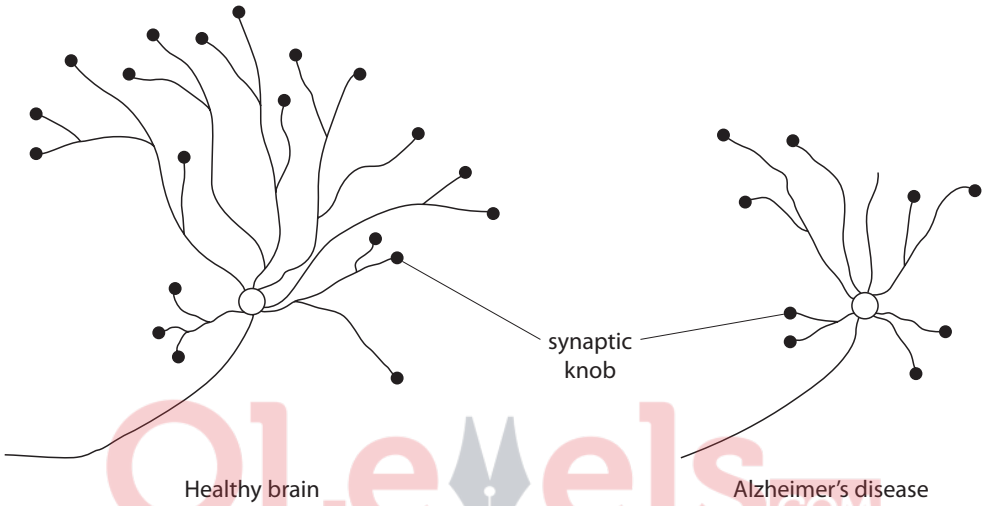


Diagram 2

(i) Give three differences between the two neurones. **(3)**

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2 .....

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3 .....

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4. Question 5 · part c.iii · 3 marks

November 2020 Paper 1

Further practice · Attempt part c.iii only.

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(ii) Acetylcholine is a neurotransmitter.

Explain the importance of acetylcholine in the functioning of the brain.

(3)

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(iii) One symptom of Alzheimer's disease is that less acetylcholine is produced in the brain.

Explain why there is less acetylcholine produced in people with Alzheimer's disease.

Use information from diagram 2 to help your answer.

(3)

(Total for Question 5 = 15 marks)

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4. Question 5 · part c.iii · 3 marks · Working and answer

Working and answer

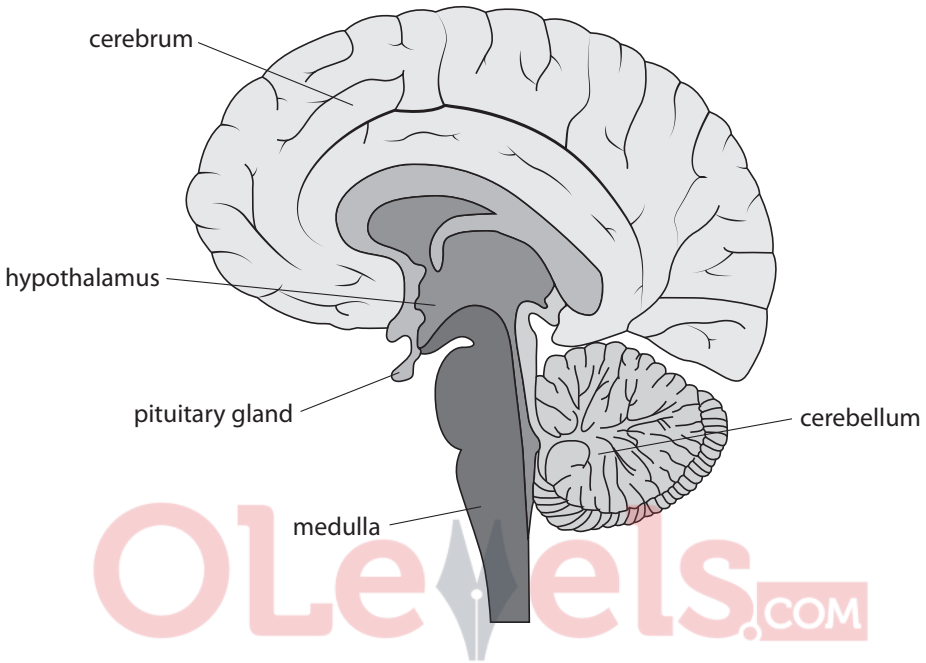
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5. Question 7 · part b · 3 marks

January 2020 Paper 2

Further practice · Attempt part b only.

7 The diagram shows parts of the human brain.



(a) (i) The table lists parts of the brain and some functions of the brain.

Complete the table by putting a tick (✓) to show if the function is correct for each part.

(3)

Part	Function	Tick
medulla	controls heart rate and breathing	
pituitary gland	involved in language	
hypothalamus	responsible for osmoregulation	
cerebellum	controls voluntary movements	

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P 5 9 9 1 0 A 0 1 8 2 4

5. Question 7 · part b · 3 marks

January 2020 Paper 2

Further practice · Attempt part b only.

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(ii) The cerebrum is divided into the left and right cerebral hemispheres.

Each hemisphere contains different areas. One of these is the motor area.

State the function of the motor area.

(1)

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(b) Nerve cells deep inside the brain produce the neurotransmitter dopamine.

Parkinson's disease is a brain disorder that destroys these nerve cells.

Explain why the destruction of these nerve cells causes the symptoms associated with Parkinson's disease.

(3)

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(Total for Question 7 = 7 marks)

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5. Question 7 · part b · 3 marks · Working and answer

Working and answer

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6. Question 5 · part a.iii · 2 marks

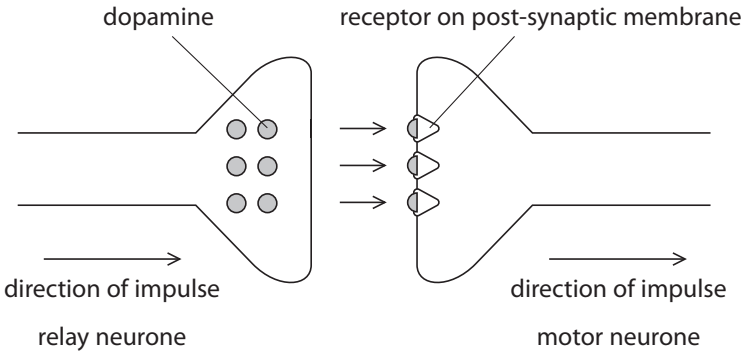
June 2023 Paper 2R

Further practice · Attempt part a.iii only.

5 (a) Dopamine is a neurotransmitter produced by nerve cells in the brain.

The nerve cells that produce dopamine are involved in stimulating movement and increasing heart rate.

The diagram shows how dopamine travels from a relay neurone to a motor neurone in the brain.



(i) Describe what is meant by a neurotransmitter.

(2)

(ii) Describe how information is transmitted from the relay neurone to the motor neurone.

(2)

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P 7 2 5 8 8 A 0 1 4 2 4

6. Question 5 · part a.iii · 2 marks

June 2023 Paper 2R

Further practice · Attempt part a.iii only.

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(iii) Suggest how an excess of dopamine will affect bodily functions.

(2)

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(b) In 2017 in the United Kingdom, 145 000 people were diagnosed with Parkinson's disease.

This figure is expected to have an 18% increase by 2025.

(i) Give two symptoms of Parkinson's disease.

(2)

1 .....

.....

2 .....

.....

(ii) Calculate the number of people expected to be diagnosed with Parkinson's disease in 2025.

(2)

number of people = ...

(Total for Question 5 = 10 marks)

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6. Question 5 · part a.iii · 2 marks · Working and answer

Working and answer

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7. Question 3 · part e · 3 marks

June 2024 Paper 2

Further practice · Attempt part e only.

3 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

People make lifestyle choices that can sometimes have a negative effect on their health. People who drink too much alcohol are at risk of brain and liver damage, certain types of cancer and mental illness. Liver damage can affect the digestion of fat. Alcohol can also affect the eyes, increasing the risk of developing cataracts that cause the lenses in the eyes to become cloudy. Alcohol is also a depressant. It increases the activity of a chemical in parts of the brain that control balance, speech, motor skills and memory. In addition, alcohol suppresses an area of the brain called the amygdala. The amygdala is responsible for the fight or flight function which prepares the body for action, initiating the release of stress hormones. These hormones increase glucose usage by the brain and other parts of the body.

When a person stops drinking alcohol after years of drinking a lot of alcohol, the amygdala in the brain becomes fully active again, which is the cause of all forms of alcohol withdrawal. Over time, if a person regularly drinks heavily, the central nervous system gets used to the suppressing effect of the alcohol, which means the brain and body are affected if the alcohol level suddenly drops. An individual can go straight into fight or flight mode as the alcohol leaves the system and levels of stress hormones begin to return to normal. Symptoms of withdrawal can be as severe as seizures and hallucinations or less severe symptoms such as loss of appetite and insomnia (an inability to sleep).

(a) Drinking alcohol and diet are two lifestyle choices.

Describe how a lifestyle choice, other than drinking alcohol and diet, can have a harmful effect on health (Lines 1 and 2).

(2)

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7. Question 3 · part e · 3 marks

June 2024 Paper 2

Further practice · Attempt part e only.

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(b) Explain how drinking alcohol affects the digestion of fat (Lines 3 to 4).

(2)

(c) State how cataracts affect vision (Lines 4 to 5).

(1)

(d) Name a part of the brain, other than the amygdala, that is affected by alcohol (Lines 5 to 7).

(1)

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(e) Explain how alcohol acts as a depressant on the nervous system (Line 5).

(3)

7. Question 3 · part e · 3 marks

June 2024 Paper 2

Further practice · Attempt part e only.

(f) After a person stops drinking alcohol, the amygdala becomes fully active again.

The release of stress hormones, such as adrenaline, returns to normal.

Give two effects of adrenaline on the body.

(2)

1 .....

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2 .....

.....

(Total for Question 3 = 11 marks)

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Working and answer